

Stroke Prediction Using Machine Learning Classification Techniques

Abstract

Objective of the Study:

The goal of this report is to predict a categorical target variable using classification techniques, specifically to determine whether a patient is likely to experience a stroke based on medical and demographic factors.

Data Description:

One morning, a dataset appeared - pulled straight from Kaggle - built for spotting stroke risks. It holds 5,110 entries, each shaped by 11 clues plus a yes-or-no outcome. Because strokes weigh heavily on global health, working with this data quietly supports UN goal number three. Prevention becomes possible when signals show up early, feeding smarter choices in real clinics. Suddenly, numbers start mattering beyond screens.

Proposed Methodology:

Starting off, the process runs through initial data checks to spot patterns. After that comes cleaning and shaping the data for use. A Neural Network takes shape alongside two older-style models - one based on probabilities, another built from many decision trees. Instead of rushing ahead, time goes into tuning settings carefully with repeated testing. Hidden within one model, a technique picks out the most useful variables. In the end, all trained systems face side-by-side review.

Key Findings:

Getting the right call mattered - Logistic Regression stood out by catching more actual stroke cases than others. Precision, recall, accuracy, plus the F1-score shaped how each model was judged. Because missed diagnoses

carry weight, high recall gave one method a cl
data like this, spotting true positives becomes key - and that is where

Logistic Regression pulled ahead. Other models tried, yet none balanced detection and reliability quite like it did.

Overall Summary:

The final model demonstrated effective stroke detection capability. Key insights include the importance of handling class imbalance and selecting appropriate evaluation metrics beyond accuracy.

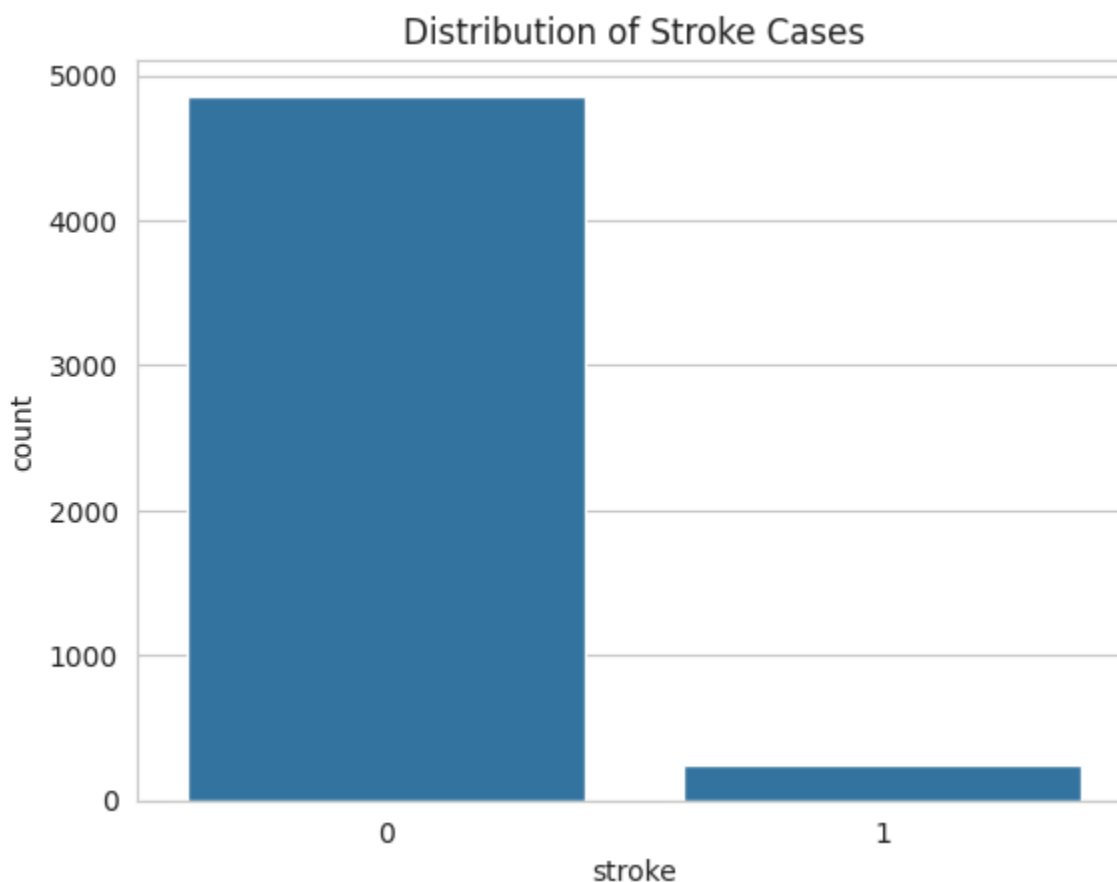
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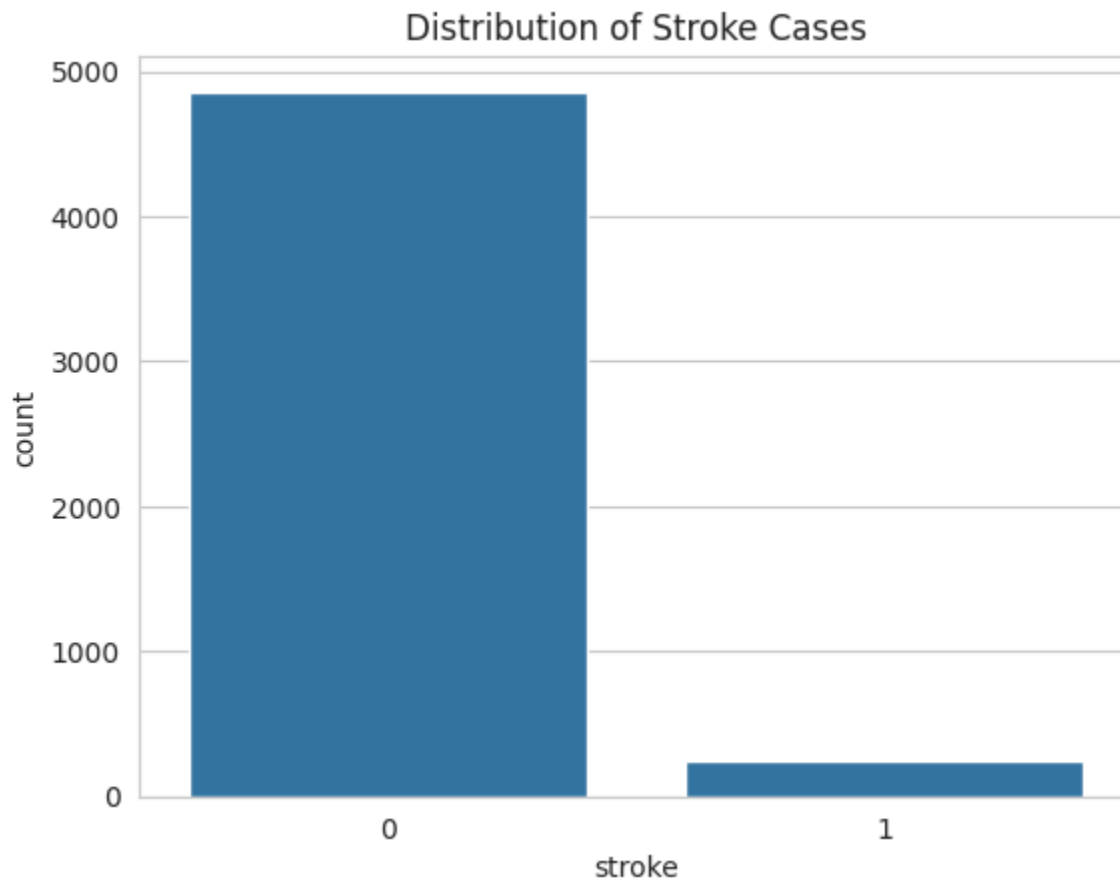
Figure List

Figure 1: Distribution of Stroke Cases



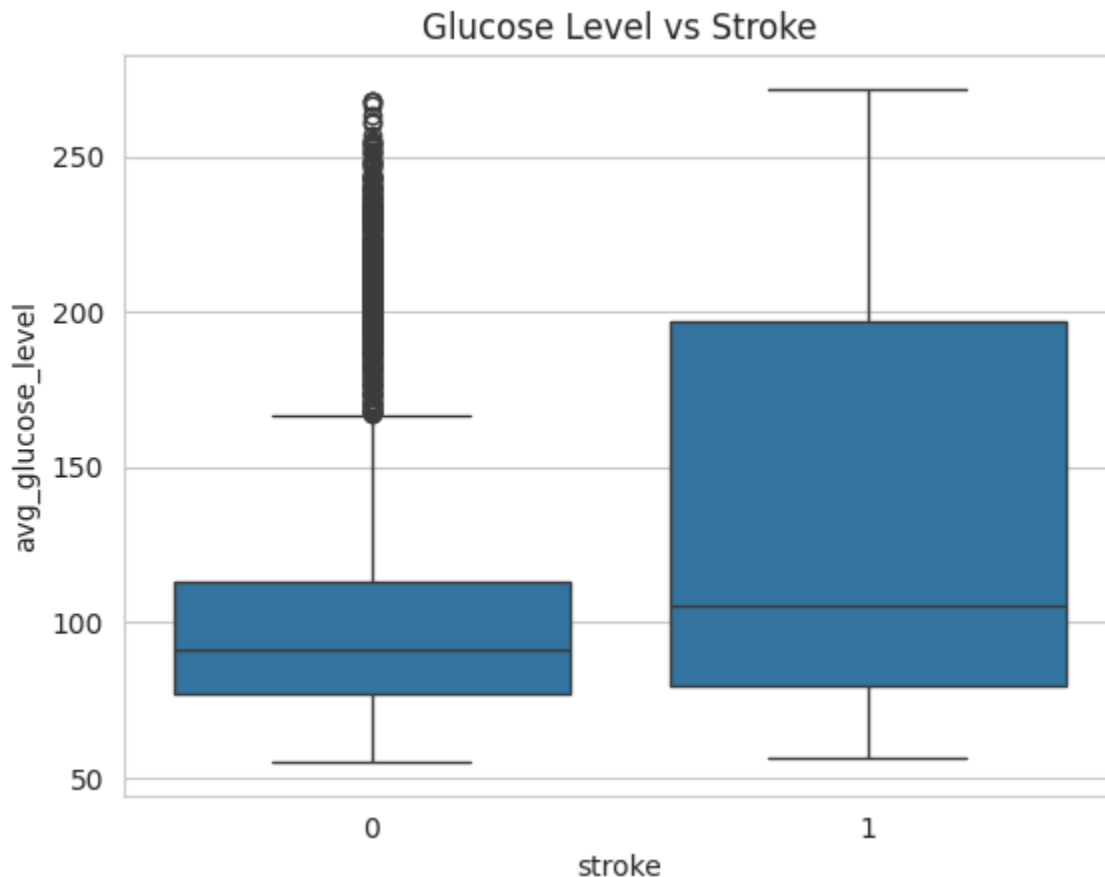
The dataset is highly imbalanced: the majority of records are non-stroke (4861) while stroke cases are only 249.

Figure 2: Age vs Stroke Distribution



Stroke cases are concentrated in older ages — the median age for stroke is much higher than for non-stroke, with fewer young stroke cases.

Figure 3: Average Glucose Level vs Stroke



Patients with stroke have a noticeably higher median and a much wider spread of average glucose levels compared to non-stroke patients.

1. Background and Overview

1.1 Problem Overview

One main aim here? Figuring out a category label through known methods in machine learning. Here though, it's about guessing if someone had a stroke - using details like age, habits, medical background. Death and lasting harm often come from strokes across the globe, so spotting risks sooner matters quite a bit.

1.2 Data Description

One way to look at it - the data comes from Kaggle, built by someone in their community, focused on stroke prediction. Patient details sit inside: things like how old they are, if they're male or female, blood sugar numbers, BMI scores, whether they smoke, along with past health issues tagged into each record. Each row tells a separate story through these markers. Found this collection already put together, no extra steps needed before diving in.

Aiming at better health outcomes, this data ties into UN goal number three by spotlighting risks before they become serious. Hidden patterns in wellness clues show where help might be needed sooner rather than later.

1.3 Study Objectives

Aiming to spot stroke risks more clearly, this work pieces together prediction tools using patient details, working around gaps in records as well as uneven case numbers. Built not just to sort data but to handle real-world hiccups like incomplete forms and rare outcomes across groups.

2. Proposed Methodology

2.1 Data Preparation and Cleaning

First up, missing numbers in the BMI data got filled in with the middle value from the group. Then came removal of the ID column since it did not help predictions at all. Moving on, categories turned into separate columns so machines could read them properly. After that, number-based fields were resized to balance their impact across models. Steps like these shaped the data before any actual modeling began

From that point onward, the data got divided into two parts - training plus test - by pulling samples carefully so each group kept the original mix of classes alive. One part stayed hidden while the model learned from the rest, making sure outcomes weren't skewed by uneven splits.

2.2 Exploratory Analysis of Data

Looking at the data started with checking its shape and makeup. Through pictures like bars and boxes, patterns began to show. Older people showed more signs of strokes. Instead of balance, one group stood out far more - those who had strokes were much fewer. Glucose values climbed higher in that same group. What emerged was unevenness, age links, and sugar levels tied together.

2.3 Model Development

Task 1 – Neural Network

A Multi-Layer Perceptron (MLP) classifier was implemented as the neural network model.

- **Architecture:** Two hidden layers with 64 and 32 neurons
- **Activation Function:** ReLU
- **Loss Function:** Cross-entropy (log loss)
- **Optimizer:** Adam

The neural network was trained on scaled features and evaluated using multiple classification metrics.

Task 2 – Two Classical Machine Learning Models

Two classical classification models were implemented:

- **Logistic Regression**
- **Random Forest Classifier**

For each model, the dataset was split into training and testing sets, the model was trained using class-weight balancing to handle imbalance, and predictions were generated for evaluation.

2.4 Performance Evaluation

Model performance was evaluated using the following metrics:

- Accuracy
- Precision
- Recall
- F1-Score

These metrics were selected because they are suitable for evaluating classification models with imbalanced classes, where accuracy alone can be misleading.

2.5 Hyper-parameter Optimization

Hyperparameter optimization was conducted for both classical models using **GridSearchCV** with **Stratified K-Fold Cross-Validation**. The F1-score was used as the scoring metric to prioritize balanced performance on the minority stroke class. Optimal parameters were identified for each model to improve generalization.

2.6 Feature Selection

Feature selection was performed using an **embedded method (SelectFromModel)** based on Logistic Regression coefficients. Features with importance values above the mean threshold were selected. This approach reduced dimensionality and improved model interpretability while maintaining predictive performance.

3. Results and Discussion

Table 4: Comparison of Final Classification Models

Model	Features Used	CV Score	Accuracy	Precision	Recall	F1-Score
Logistic Regression	Selected Features	0.234	0.76	0.14	0.80	0.24
Random Forest	Selected Features	0.196	0.95	0.20	0.40	0.27

3.1 Summary of Key Findings

The evaluation results show that Logistic Regression achieved significantly higher recall for stroke cases compared to Random Forest. Although

Random Forest produced higher accuracy, it failed to consistently identify the minority class.

3.2 Final Model Selection

Based on recall and F1-score, **Logistic Regression** was selected as the final model. It demonstrated superior ability to detect stroke cases, which is critical in healthcare applications.

3.3 Implementation Challenge

Several challenges were encountered, including severe class imbalance, missing values in medical attributes, and the misleading nature of accuracy as an evaluation metric for this dataset.

3.4 Scope for Future Work

Future improvements could include applying resampling techniques such as SMOTE, exploring cost-sensitive learning, or incorporating additional clinical features to improve stroke detection.

4. Analytical Discussion

4.1 Evaluation of Model Performance

The models were evaluated using Accuracy, Precision, Recall, and F1-score. Logistic Regression performed better in identifying stroke cases, while Random Forest was biased toward the majority class.

4.2 Effect of Hyperparameter Tuning and Feature Selection

Hyperparameter tuning improved model generalization, particularly for Logistic Regression. Feature selection reduced noise and helped the models focus on the most relevant predictors, resulting in more stable performance.

4.3 Result Interpretation

The results align with expectations for imbalanced medical datasets, where simpler and well-regularized models often outperform complex ones.

4.4 Study Limitations

Limitations include dataset imbalance, limited feature diversity, and reliance on a single dataset. Neural network performance was also constrained due to limited imbalance-handling capabilities.

4.5 Recommendations for Future Research

Future research could investigate advanced ensemble methods, deep learning architectures with imbalance-aware loss functions, and larger multi-source datasets.

5. References

- Stroke Detection Dataset (Kaggle)
- United Nations Sustainable Development Goals (SD3)